Farming for Soil Health

Best practices for managing Irish agricultural soils

Fiona Brennan, Dermot Forristal, Veronica Nyhan, Luis Lopez-Sangil, Saorla Kavanagh, Cathal Somers, Owen Fenton, Giulia Bondi, Karen Daly, Lilian O'Sullivan, John Finn, Aaron Fox and David Wall.

To be a successful farmer one must first know the nature of the soil."

-Xenophon [Oeconomicus, 400 B.C]

We know more about the movement of celestial bodies than about the soil underfoot"

-Leonardo da Vinci

Our soils must be nurtured and protected"

- Michael D. Higgins (Uachtarán na hÉireann) 2021

Out of the long list of nature's gifts to man, none is perhaps so utterly essential to human life as soil"

- Hugh Hammond Bennett

The nation that destroys its soil, destroys itself".

–Franklin D. Roosevelt

Essentially, all life depends upon the soil"

— Charles Kellogg

While the farmer holds the title to the land, actually it belongs to all people because civilisation itself rests upon the soil".

–Thomas Jefferson

Soil health is our wealth

Soil is the natural medium on the Earth's surface that supports plant growth and sustains life by providing nutrients, water, a structure, and a habitat for organisms. It is a dynamic, biologically active ecosystem composed of a mixture of mineral and organic materials, air, and water. Soil is one of the most valuable resources on the farm. It takes a long time to form, can be lost or damaged quickly, and if it becomes badly degraded, it's not easy to fix. Farmers are the main custodians of our nation's soils – how we manage our soils today impacts not only this year's harvest, but harvests for many years to come. Good soil health is amongst the greatest legacies we can give to society, both now and into the future. Soils are at the heart of our food systems and are essential for environmental and planetary health. They have a direct impact on farm productivity, input costs, long-term resilience, and the well-being of both local and global communities that depend on them. As we face more extreme and unpredictable weather patterns, healthy soils are more resilient and less likely to be negatively affected by climatic events. Good soil health benefits us all!

Healthy Soils

Healthy soils are in good chemical, biological, and physical condition so that they can provide ecosystem services that are vital to humans and the environment, such as safe, nutritious and sufficient food, biomass, clean water, nutrient cycling, carbon storage and a habitat for biodiversity[1]

Balancing what soil does for us.

A soil's natural properties determine what it can do. How we manage it affects the balance of these functions. Some benefits go hand-in-hand: for example, increasing soil organic matter can boost carbon storage, water retention, and biodiversity, positively affecting multiple functions. But there can also be trade-offs – like a seesaw: putting a lot of weight on one function may reduce the effectiveness of others. For example, drainage of a peat soil may enhance productivity, but that soil may hold less water and carbon. Different farms aim for different balances, but good management should always protect soil health and optimise its functions.

Different soils need to be managed in different ways

Agricultural soils are managed to produce food, feed, and fibre; but when they are well-managed and healthy they also help protect, purify and store our waters, provide a home for biodiversity and play a major role in regulating climate change - among many other benefits. In other words, soils are multifunctional! Did you know, for example, that many of our most important modern antibiotics originally come from soil organisms? The benefits of soils are far-reaching! But when soil is damaged, it can't deliver this multifunctionality as well. **For example in one Irish study, compaction that reduced the soil's pore space by 12% led to about a 20% drop[2] in barley yields: the damage reducing the soil's ability to hold water and air, and to support healthy plant growth.** When it comes to soil management, one size does not fit all! Management must be tailored to the local context, considering the soil type, the environment, the landscape, the farming system and drawing on local knowledge. **However, there are some general principles for good soil management that can be applied to every farming system to help keep soils healthy.**

Purpose of this handbook

The aim of this handbook is to share practical, evidence-based principles on soil health that farmers, growers, and advisors can use to manage their soils more effectively. We also hope it's a helpful resource for those involved in soils education or those shaping soil-related policy or legislation. Like any general advice, these principles work best when farmers or land managers adapt them to suit their own farming system and local conditions. Most recommendations in the principles are either cost-neutral or cost-effective. However, in some cases (and within some systems), there may be agronomic trade-offs that farmers should consider when choosing what practices to implement.

Starting a soil health journey can feel daunting, and it's not always easy to know where to begin. Take small steps and learn as you go. Start with the easy wins — once you see success, you'll feel more confident tackling the bigger challenges. And remember, you can always reach out for help along the way.

Principles of good soil management

PRINCIPLE 1

Know your Soils

Why?

All soils have unique properties. Ireland has a huge variety of soils, with 213 different soil types (series; see section on soils of Ireland; page 26)[3]. It is common to have multiple soil types on a farm (or even within a field!), each of which may need to be managed in a different way. The type of soil may also have different features depending on its position within the landscape. **Understanding what kind of soil you have can help you make better decisions about how to manage it more effectively and sustainably.**

What to do?

- **Get to know what soils** you have on your farm, and where they are located. When planning how to manage your soils, think about their place in the landscape – including slope, elevation, bedrock, proximity to water sources and local climatic conditions.

Soil texture

Soil texture is the share of sand, silt and clay in your soil. Feeling and working a small amount of soil in your hand (hand texturing) can help identify what your soil texture is[4]. Gritty soils feel sandy, clay soils feel sticky, silty soils feel smooth and buttery, while loam soils are a balanced mixture of all. Soil texture is largely determined by the parent material (the rock or sediment from which it formed) and natural weathering processes. Unlike soil structure, which can be altered through management, texture usually doesn't change.

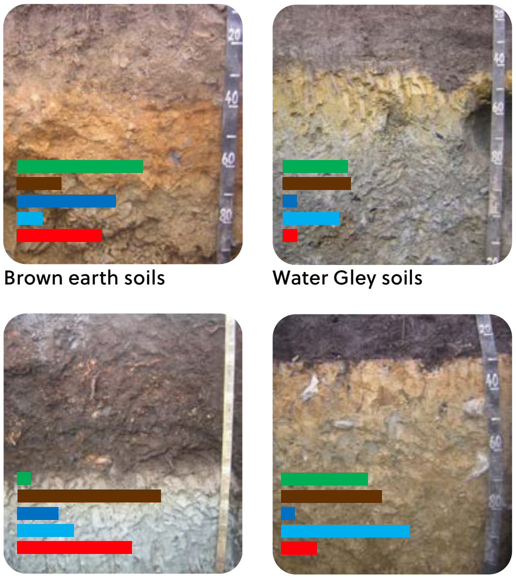
Ask yourself:

- What are the key soil characteristics – such as its texture, drainage capacity, depth, pH or organic matter content?
- Is your soil mineral or organic?Organic soils contain ≥20% soil organic carbon (SOC) in the top 30-40cm, which roughly corresponds to ≥30–40% soil organic matter (SOM); they are usually very dark in colour.
- Is your soil clayey and prone to waterlogging, or is it sandy and free-draining, or somewhere in between?
- Are there areas that are vulnerable to problems like compaction or poaching, water stagnation, soil erosion, or loss of organic matter?
- Are there watercourses or sensitive habitats nearby that could be affected by how you manage the soil?

Digging for Knowledge

Going out with a spade and looking at the soil for yourself can be the most informative way to get to know it and can save time and money in the long run. If necessary, dig some shallow soil pits (40-50cm) to explore your soil, looking out for soil features (for example, changes in texture, or mottles (blotches of orange or blue/grey that indicate poor drainage or fluctuating water tables). Consider teaming up with a farmer friend or discussion group to dig in each other's fields— you'll learn a lot more, and it's more fun.

- **Obtain local knowledge on past management or soil properties** (such as fields prone to flooding or previous soil test results) where available, as such information can be invaluable for guiding future management. Note fields that have previously been drained with open drainage ditches, in-field pipes, or mole/gravel mole drains. Their presence identifies fields that have a history of poor drainage.
- **Access free national maps** that show soil types, subsoils (the layer of soil located beneath the topsoil), slopes, bedrock, and pollution risks[6,7]. Identifying the soil association where you are located will give you a general idea of the main soils likely to be found in your area. Additionally, older ordinance survey maps can be useful as they often contain previous field divisions. European maps, like those from the LUCAS survey, are also available[8]. But keep in mind that most of these maps are too coarse a resolution to show what's really happening at farm or field level. It is recommended to always check the land yourself (for example with a few soil pits or local knowledge) before using these maps to guide decisions on soil management. Long-term weather data (available from Met Éireann[9]) can also be useful for looking at climatic trends.
- If you haven't had your soil fertility tested recently **consider conducting some baseline chemical analysis** to help you tailor management practices effectively (Principle 6).



Brown earth soils

Water Gley soils

(Blanket) Peat

(Histic) Luvisols

Fig. 1
Different soil types have different potential capabilities and vulnerabilities

Legend: potential soil functions and vulnerabilities
- Plant productivity
- Carbon sequestration & storage
- Water percolation
- Nutrient retention
- Resilience against compaction

Playing to our soils' strengths

All soils have their own properties that result in different capabilities and vulnerabilities[5] (Fig. 1). For example, soils with a high clay content can retain more nutrients and soil moisture, and sequester more organic carbon, but they can be slow to drain and are more susceptible to water stagnation and compaction. Sandy soils, on the other hand, drain quickly and are more resilient to compaction, but they don't hold on to nutrients, water and carbon as effectively. The key is to manage the soil you have, playing to its strengths!

PRINCIPLE 2

Assess your soil health

Why?

Checking the biological, chemical, and physical condition of your soil (Fig. 2) can help you make better management decisions and pinpoint, address and resolve any soil health issues on your farm[4]. It also provides valuable insights into the capacity of soils to function, for example, to produce healthy crops, to act as a habitat for soil life, to cycle nutrients, and to store water and carbon (Fig. 3). Most of us would give our cars or ourselves a regular check-up – why not our soils?

Poor soil structure can hinder crop establishment, compromise plant health and yield, increase vulnerability to erosion, reduce water infiltration and storage, and disrupt nutrient cycling. Chemical assessment can help you to manage soils in a way that enhances nutrient use efficiency, reducing costs associated with chemical fertiliser inputs and decreasing nutrient losses to the environment. Healthy soils are living soils; ensuring you have a healthy soil biota is crucial for making sure your soils function efficiently.

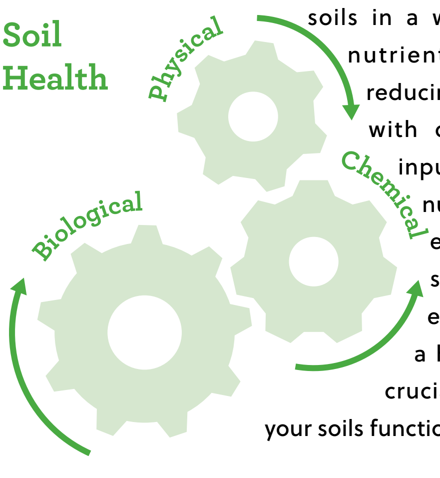


Fig. 2 A soils capacity to function is based on the interaction of its biological, chemical, and physical properties

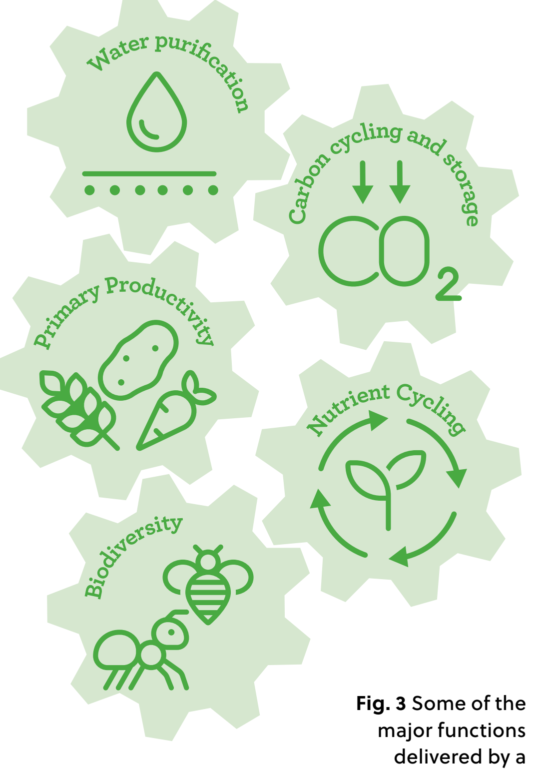


Fig. 3 Some of the major functions delivered by a healthy soil

What to do?

- **Look at the best and worst areas on your farm.** For the latter pay particular attention to areas on your farm where you think there might be soil health problems. Are there areas where you have evidence of poor crop growth, poor root development, reduced water infiltration, compaction, reduced organic manure decay rates, soil erosion, or that you can observe surface damage caused by machinery or animal traffic? These are areas you might want to investigate further to identify specific problems and their solutions. Check if variations in crop or grass growth are associated with traffic patterns indicating soil damage – become your own detective!
- **Use visual assessment techniques to evaluate your soil physical health** by observing soil features such as colour (darker colours indicate more soil organic matter), soil structure (size and shape of aggregates), rooting depth, porosity, and biological activity (earthworm channels, the presence of organisms)[2]. Look for evidence of compaction (dense sections of soil) and waterlogging (grey or orange mottling) to diagnose issues (page 11). Visual Evaluation of Soil Structure (VESS) guides exist for grassland soils (GrassVESS[10]) and arable soils (Double Spade[2]) that enable you to score the health of your soil quickly and easily. They require limited equipment (a spade, a trowel and a measuring tape) and have no cost other than your time. Carry out a number of VESS assessments on contrasting areas of your farm (including your problem areas) to see how your scores compare under different management. Repeating the VESS will allow you to see how your soils are changing over time. Consider also examining the soil profile to a depth of 40–50 cm to identify potential subsoil problems. Sometimes topsoil can look fine even when issues like compaction or low-permeability layers exist below.

Soil compaction

Soil compaction occurs when soil particles are compressed together, making the soil denser and reducing the size and number of pore spaces between them. Signs of compaction often include poor crop growth, surface ponding or waterlogging due to low infiltration, poor root growth, hard angular aggregates, limited pore space and low biological activity.

- **Assess chemical health by conducting regular chemical soil analysis tests**, particularly for available macronutrients and pH.
- **Visually assess your soils for living organisms and indicators of biological activity.** Soil life is rich and varied, and much of it is too small to see with the naked eye. To measure the full range of soil biodiversity, specialist lab tests are usually needed, and these aren't always easily available to farmers. But the good news is that you can still learn a lot about the biological health of your soil just by looking closely and thinking of the soil as a living habitat. Ask yourself:
    - Is the soil structure good? Is there space for biodiversity?
    - Is there plenty of organic matter and roots to feed soil life?
    - When you dig a bit, do you see larger soil creatures like earthworms, ants, beetles, woodlice, centipedes, or millipedes?
    - Can you spot signs of activity, like earthworm casts or the fast breakdown of manure?

For larger organisms like earthworms and pollinators that live in soil, you can count them during seasons when they are active[11,12].

Soil structure

is the architecture of the soil—how its different components (sand, silt, clay, organic matter, and pore space) are arranged. The building blocks of soil structure are tiny, naturally formed clumps called aggregates, which can stick together to form larger aggregates. The cracks, gaps, and spaces within and between these aggregates are called soil pores, which can be filled with air or water. How your soil is structured really matters to how it can function. Good soil structure lets water, air, and roots move freely and provides a home for soil life, supporting healthy crops and environmental health[2]

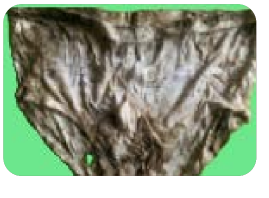
Try comparing numbers across fields with different managements to see how practices affect soil life. You can also use simple functional tests — like the cotton underwear decomposition test[13] — to check and compare biological activity.

Fig. 4 Score Level of decomposition

1
75-100%
Nearly all decomposed

2
51-74%
Over half decomposed with some remaining

3
26-50%
Some decomposition, at least half remaining



4
0-25% Little to no decomposition

Good Health indicator

- **Round Aggregates** (soil clumps) that are predominately small is a sign of good physical health.
- **Deep root growth** - unrestricted root growth through the soil layers and aggregates is a sign that soil is loose and well-aerated, allowing plants to access water and nutrients more easily.
- **Soil Fauna** – spotting earthworms and other organisms, or evidence of their presence, shows active life in the soil.
- **Soil organic matter**. Darker colours indicate high soil organic matter.
- **Bio channels** – Channels left by roots or organisms are an indication of air and water movement through the soil, and good biological health.

Good health indicators


Round aggregates
(small clumps)


Deep root growth


Soil fauna


Soil organic matter
(darker colours)


Bio Channels
(roots & fauna)

Poor Health indicator

- **Plough pan** – A hard, dense and compacted layer just below the plough depth that restricts root growth and water movement.
- **Compaction** – Soil is compressed and hardened, reducing air spaces and making it harder for roots and organisms to thrive.
- **Limited biology** – Few signs of earthworms or other living organisms signal poor biological health.
- **Mottles** – Grey/blue or orange blotches show poor drainage and fluctuating oxygen levels in the soil. Grey areas show where oxygen is lacking due to waterlogging, and iron is reduced. Orange blotches are where iron has re-oxidised when the soil encounters oxygen again.
- **Poor structure** – Large and angular aggregates or dense soil that is hard to break in your hand is a sign of poor physical health.
- **Water stagnation** – Pooled or standing water days after rainfall indicates drainage problems and lack of oxygen in the root zone.

Poor health indicators


Plough pan
(25 - 30cm depth)


Compaction
(traffic or poaching)


No biology
(no root growth)


Fe redox mottles


Poor structure
(no round aggregates)


Water stagnation
(after rainfall)

PRINCIPLE 3

Avoid physical damage

Why?

When soil is physically damaged, its structure gets disrupted. This damage can reduce soil resilience and its ability to support crops - whether arable, horticultural or grassland – and can also harm the environment[2,14]. Where soils are severely damaged this can result in:

- Poor crop establishment, lower yields and reduced plant health
- Fewer beneficial soil organisms and reduced biodiversity
- Slower water infiltration and poorer water storage in the soil
- More surface runoff, in some cases to waters, leading to water pollution
- Loss of soil fertility
- Less carbon stored in soils, and increased nitrous oxide emissions
- Increased vulnerability to erosion and compaction
- Reduced resilience to droughts and heavy rains
- Issues for existing installed drainage systems

Physical Soil Damage

The most common type of physical soil damage in Ireland is compaction. This usually happens due to heavy machinery or livestock trafficking, especially when soils are wet. Poorly drained soils are more at risk of severe damage. Erosion — the loss of topsoil by water, wind, deforestation or tillage — can be a problem in certain areas and is likely to become more common with more frequent extreme weather events.

What to do?

Prevention is better than cure!

Avoid damage to soil in the following ways:

- For both arable and grassland soils, **reduce heavy machinery traffic** on soils as much as possible
- **Avoid or limit machinery or animal traffic on wet soils**. Soil structure is more vulnerable when soils are wet. Always check if the soil is sufficiently dry before starting cultivations. This requires getting the right balance between soil being dry enough to work without risking late crop establishment. Check soil conditions early in the year before any machinery operations, when the risk of long-term subsoil compaction is higher. Be particularly careful in advance of spring soil tillage operations, where the soil may be dry at the surface but still vulnerable to damage at depth. Consult the Agricultural Services webpage on the Met Éireann website or mobile app for up-to-date forecasts, risks and warnings including information on soil water deficits - the amount of rain needed to bring the soil moisture content back to field capacity- which can help determine when it is safe to traffic the soil.

However, field examination of the soil is essential and remains the primary determinant of when it is safe to work or traffic.

- In grazed grasslands, **avoid animal poaching where possible.** Using an on/off grazing system can help protect the soil during the shoulders of the year when it is most vulnerable to damage from animal treading. Avoid overstocking and rotate livestock regularly to reduce damage to soil structure.

- **Lighten the load!** Machinery is getting heavier, which might help with labour efficiency — but it's tough on your soil. 'Fully mounted' equipment today can lead to axle loads of 8 to 13 tonnes when turning, putting serious pressure on soil. To reduce compaction:
  - Use lighter machinery where possible.
  - Spread weight by using larger/wider tyres, or more wheels/tyres, allowing lower tyre pressures and reduced ground pressure. Use newer tyre technology such as Very High Flexion (VF) tyres and central tyre inflation systems (CTIS), which allow more deflection and greater contact area provided the correct inflation pressure is used. Check your axle loads on a weigh bridge and check the tyre manufacturer's charts (available online) and, whenever it is practical, adjust your pressure to the load and speed. You may need to readjust tyre pressure for roadwork.
  - Reduce axle loads by using trailed or semi-mounted equipment machinery instead of fully-mounted where appropriate.
- When reseeding grassland or sowing arable crops **consider reducing tillage depth, frequency, and intensity.** This can improve soil strength, making fields easier to travel on, and supports soil organisms that help maintain good structure. However, in a wet climate like Ireland's, this approach needs careful management — compaction can still happen and may occasionally require adjusting of tillage depth or intensity. There may also be trade-offs between reduced tillage and challenging weed issues. Deep-rooted cover crops or grassland species such as chicory can also improve soil structure and reduce the need for mechanical tillage, but in some soils they are unlikely to help significantly if the soil is already severely compacted. In summary, do as little tillage as possible and as much as necessary - as soil structure improves it needs less work and expense to produce a seedbed.

- Where cultivating, **vary cultivation depth** occasionally to avoid forming compacted layers (such as plough pans) at the same soil depth. Include crops with different rooting depths in your rotation to naturally break up minor compaction and add organic matter deeper in the soil — this is especially important in zero-till systems.
- **Pay extra attention to headlands,** as they are more prone to compaction and damage from frequent machinery traffic.

Plan the sequence of operations carefully — where possible, cultivate and sow the main field first, and do headlands last. Consider grassed headlands for turning or rotate turning areas to different parts of the field to allow original headland areas to recover. If deeper tillage is needed to fix compaction, use it cautiously, and protect the area afterward by reducing traffic and carefully matching tillage depth and intensity to the area's requirements.

- **Use fixed tramlines and other controlled traffic farming (CTF) measures** to reduce machinery damage in the wider area, even in dry conditions.
- **Choose sites carefully for crops that need intensive cultivation** (and destoning), such as carrots, beet, potatoes, or grazed forage crops. These crops are often harvested late in the season, when soils may be wet, increasing the risk of serious damage to soil structure if not managed well.
- **Maintain or increase soil organic matter** (see Principle 5). It improves soil elasticity, helping it resist compaction and smearing. Even small changes in organic matter levels can make a big difference to how easily soil becomes compacted. In some cases, liming can also enhance soil aggregate stability, making it less prone to damage.

- **Keep a living root system in the soil** and avoid bare soil as much as possible to enhance physically anchoring of soil. Root systems create channels and pores that enhance soil aeration, water infiltration, and aggregation and reduce erosion from wind and water. In arable systems, early-sown cover crops or natural regeneration can provide a living root between main crops. Additionally, cover crops (also known as catch crops) can 'catch' nutrients over the winter period when they are most vulnerable to be lost to the environment. Where cover crops cannot be established quickly, existing stubble from the previous crop can help retain stable soil surface conditions over-winter.
- In horticultural soils, **applying protective organic amendments** (for example, composts or bark mulches) to the surface, or incorporated into topsoil, can protect bare soil. However, be aware that some amendments may immobilise soil nutrients, making them less available for the following crop in the shorter term. Leaving crop residues can also protect soil.
- **Visually assess soil structure regularly** to identify problematic areas so management can be quickly adjusted if required.
- Where soils are particularly vulnerable to erosion, **implement measures to help slow down the movement of wind or water through the landscape** (such as wind breaks, buffer strips, cultivation direction, cultivation depth and intensity, timing of reseeding, stubble/ cover crop management, continuous plant cover).
- **Take particular care of soils after disturbances**, as such soils are more prone to subsequent compaction or loss through erosion. Do so by minimising trafficking, maintaining plant cover, diversifying plants, planting perennial plants, etc.

- **Consider the soil impact when selecting all crops in a rotation**. A fallow year may offer some benefit where soils have been badly damaged.
- **Improve design of farm infrastructure** (for example, gateways, roadways, cattle troughs) to minimise damage to soil[4]
- **Maintaining hedgerows** in fields can help with soil structural health and reduce compaction around the field margins.

Where structural damage has occurred:

- **Identify** the location and soil depth that is impacted and assess the nature and level of damage. This helps you decide what kind of action is needed. Shallow compaction (0-15 cm) can often recover naturally through root growth, the activity of soil organisms, and natural wetting-drying or freezing-thawing cycles. Deeper compaction (down to 50 cm), usually caused by heavy machinery, is harder to fix and may require more intensive intervention. In grassland systems, aerators are generally only effective when compaction is very close to the surface.

- **Maximise organic matter return to damaged areas**, particularly making use of on-farm amendments such as farmyard manure. This will improve soil structure and attract soil organisms such as earthworms that create channels as they move through the soil and improve infiltration.
- **Give damaged fields time to recover;** reduce traffic at vulnerable times (altering cropping or grazing pressures if beneficial) and allow the crop growth or other vegetation cover to supplement the restorative action of natural soil wetting-drying and freezing-thawing cycles.
- **Use mechanical loosening of the soil at depth** (for example, subsoiling) **only as a last resort** for severe subsoil compaction. First, assess the depth of the compacted layer, then target the subsoiler just below it. Subsoil only when the soil is dry to ensure the compacted soil shatters effectively rather than smearing. If possible, combine subsoiling with a fast-growing, deep-rooting cover crop or allow natural regeneration. After mechanical loosening, the soil may remain vulnerable to re-compaction, which could result in deeper structure damage that is harder to fix. Adopt a cautious soil management approach afterward—especially on wet soils—to prevent re-compaction and allow the loosened soil to stabilise.
- **Consider changing headland areas for turning machinery** to other field sides or use a second temporary headland inside the existing headland to allow damaged areas to recuperate.

PRINCIPLE 4

Support Soil Biodiversity

Why?

Soils are home to nearly 60% of all life on the planet, making them one of the largest reservoirs of biodiversity on Earth. Soil biodiversity is crucial for maintaining healthy, productive, and resilient agroecosystems[14]. Soils support many beneficial organisms such as insects (bees, hoverflies, ground beetles), arachnids (spiders), microorganisms (bacteria and fungi), and nematodes, all of which can act as natural pest enemies. Soil organisms play a fundamental role in key soil functions including nutrient cycling, pest and pathogen control, climate regulation, and water retention, and are essential for maintaining soil structure and stability[15,16]. Crops grown in healthy soils can be less prone to pest infestations and diseases, reducing the reliance on chemical controls, while also requiring fewer fertiliser inputs. Additionally, soils rich in biodiversity are more resilient to environmental changes and stresses like droughts, floods, and other disturbances.

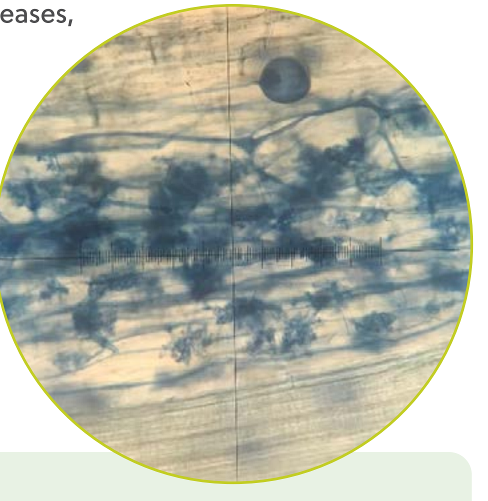


Fig. 5 This microscope image shows part of a plant root colonised by arbuscular mycorrhizal fungi. The round spores and the highly branched arbuscules — where the fungus and plant exchange nutrients — are visible. Photo credit (Julia Hess)

Mycorrhizal fungi

Did you know that most plants form partnerships with beneficial soil fungi called mycorrhizal fungi (myco = fungus, rhizo = root)? These fungi helped plants colonise land hundreds of millions of years ago. The most common type, arbuscular mycorrhizae, grow into plant roots and produce dense networks of hyphae that extend into the surrounding soil. In exchange for carbon from the plant, these networks help plants take up nutrients and water and can improve resistance to disease and drought. Avoiding over-application of phosphorus fertiliser, maintaining plant diversity and limiting the use of broad spectrum fungicides, helps maintain healthy mycorrhizal communities.

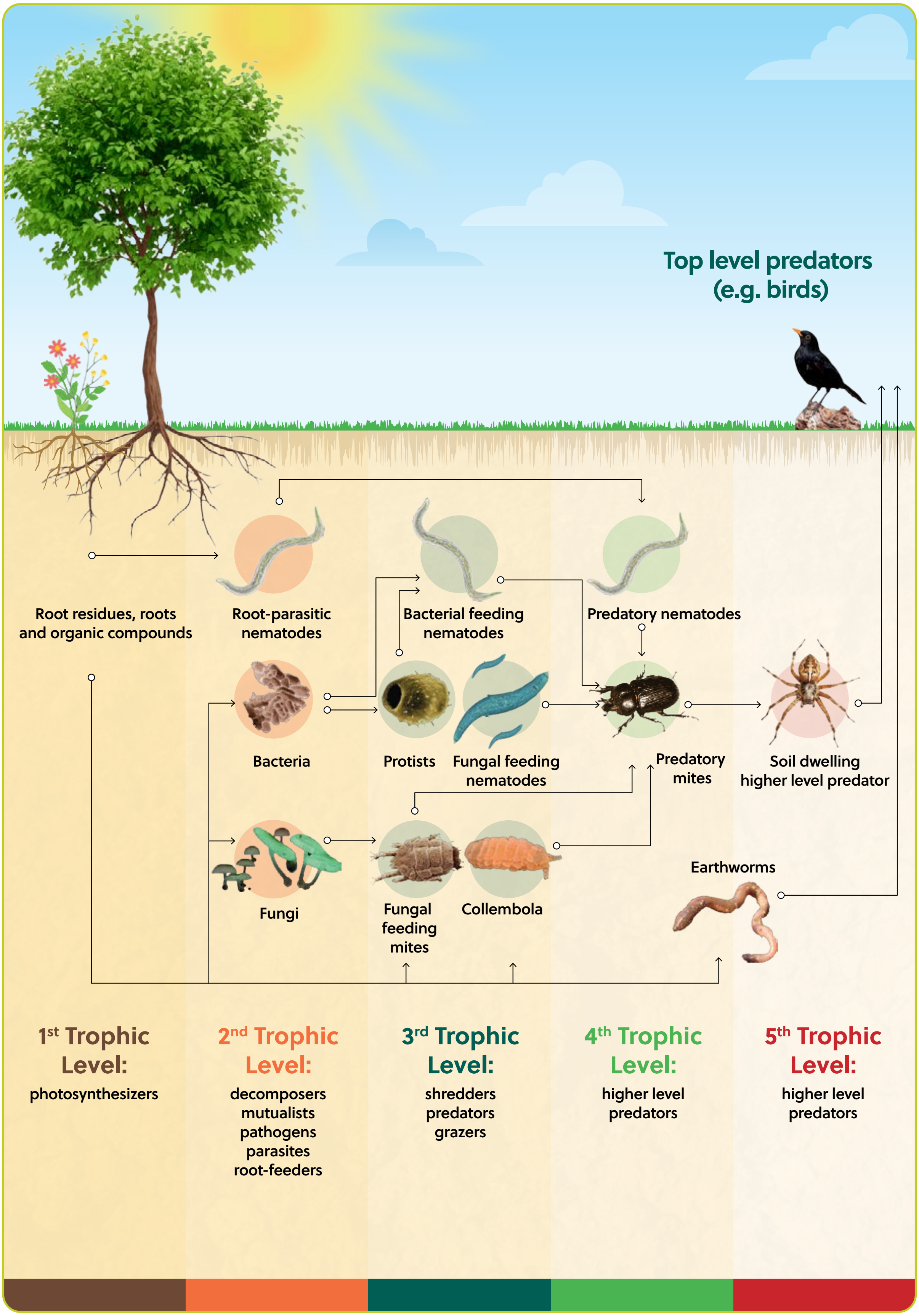


Fig. 6 The soil food web

What to do?

The following practices encourage a diverse soil biology.

- **Maintain a good soil structure**. Physical damage of soils can result in loss of habitats and food sources that support soil organisms (see Principle 3). Much of soil life resides inside the soil pore space so keeping a good soil structure is important for supporting biology.
- **Get to know what species live in the soil.** The 'Soil biodiversity of Ireland' poster is available to download at: https://teagasc.ie/wp-content/uploads/uploads/media/website/environment/soil/Soil-Biodiversity-of-Ireland.pdf
- **Maintain or build soil organic matter** (see Principle 5). Organic matter is a food source used by soil organisms and the foundation of the soil food web (Fig. 5). Soil biology also transforms nutrients bound in soil organic matter into plant available forms – making it an important plant fertiliser.

Fig. 7 Soil Biodiversity of Ireland

- **Keep a living root system in the ground as much as possible.** Living plants release simple sugars (amino acids, and organic acids) through their roots. These sugars act as a continual food source for soil microorganisms and can increase microbial activity, improving nutrient cycling, organic matter breakdown, carbon storage and soil structure. Living plant roots also form mutually beneficial (symbiotic) relationships with beneficial (mycorrhizal) fungi. Dead roots become organic matter, improving soil organic carbon levels and feeding decomposers.

- **Diversify cropping**. Avoid growing the same crop continuously (monoculture cropping), as this can reduce the variety of habitats and food sources available for soil organisms. In sown grasslands, adding legumes or herbs into the sward increases plant diversity and can improve productivity. Semi-natural grasslands already have high plant diversity, so protecting this is a priority. In arable systems, cover crops can benefit soil biodiversity, and rotating annual crops can also help break pest and disease cycles. Be mindful that careful choice of cover crop species is also needed to avoid build-up of soil borne diseases such as clubroot on brassica crops.

- **Minimise soil disturbance**. Consider reducing the frequency, depth and intensity of tillage operations, which can disrupt the habitat for soil organisms, reduce fungal biomass and negatively impact macrofauna such as earthworms. Earthworms move toward and away from food sources like cow pats, and in doing so, create channels that help improve soil structure. Irish research has shown that reduced tillage can also reduce aphid pressure on crops when compared to plough-till based systems[17]. There may be trade-offs, however, with increased weed or pest pressures potentially leading to greater pesticide use, which can negatively affect pollinators. Following a break crop such as legumes or oilseed rape is often an ideal time to establish crops with less disturbance as the soil is already friable.

Earthworms

Charles Darwin famously called earthworms 'nature's plough'. These soft-bodied invertebrates tunnel through the soil, creating channels that improve aeration and drainage while mixing organic matter into deeper layers. By breaking down leaves and crop residues and releasing nutrients, they boost soil fertility and help plants grow — making them essential allies for farmers and gardeners. Healthy earthworm populations are a simple, visible sign of good soil health. Reducing soil disturbance, increasing organic matter, and maintaining soil moisture can help increase earthworms on farmland.

- **Reduce chemical inputs.** The use of chemical inputs such as fungicides, herbicides, insecticides, and molluscicides, can have negative effects on biodiversity, and soil health. Minimising chemical inputs and maximising input efficiency is one way to help protect soil biodiversity. Consider adopting integrated pest management (IPM) approaches for controlling pests, diseases, and weeds to reduce reliance on chemical pesticides. Some examples of IPM are pest monitoring, crop rotation, machinery cleaning, and practices that support beneficial organisms—such as insects, arachnids, microorganisms, and nematodes—which act as natural enemies or enhance plant resistance to attack.

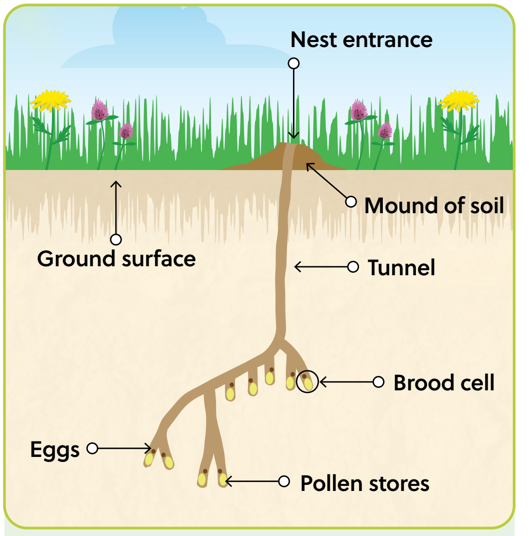


Fig. 8 Below ground solitary mining bee nest.

Ground nesting bees

Over 60% of Ireland's and Europe's bees' are ground nesting solitary bees. They nest by making tiny burrows in bare earth (clay, peat, sand and soil). Some species of ground nesting bees found on farms in Ireland are the grey (or ashy) mining bee (*Andrena cineraria*), the chocolate mining bee (*Andrena scotica*) and Gwyne's mining bee, (*Andrena bicolor*).

Biodiversity tip: Scrape back a small area of soil (150cm² to 12m²) on a suitable bank (well-drained south or south-west facing bank). More information available at: https://biodiversityireland.ie/app/uploads/2022/05/ActionSheet_Solitary-Bees-WEB-2.pdf.

- **Prioritise the use of organic fertilisers,** such as slurries and farmyard manures, which are generally more beneficial for soil organisms than inorganic fertilisers.
- **Avoid overloading soils with nutrients** (see Principle 6), as this can disrupt beneficial plant–microbe interactions, disturb soil food webs, and increase the abundance of microorganisms that produce nitrous oxide, a potent greenhouse gas[15].

- **Maintain farmland habitats**. Trees, hedges, wetlands and the wildlife that belong to them will contribute to increasing soil biodiversity in the area.

Springtails

Springtails (Collembola) are tiny (1–3 mm), ancient, soft-bodied soil arthropods with 6 legs that feed on decaying plant material, fungi and microbes, helping to break down organic matter and recycle nutrients. They are among the main biological control agents of microbial populations, helping to keep fungal and bacterial communities in balance — their presence is a sign of healthy soil. They're called springtails because many species have a forked tail like appendage (the furcula) that snaps against the ground to catapult them into the air like a tiny spring.

Dung life

Dung-loving beetles, worms, and flies are important for nutrient cycling. Anti-parasitic drugs used on livestock can have negative effects on dung loving insects and knock-on effects for the birds and mammals that eat them. Biodiversity tip: Integrate a parasite management plan into your animal management plan to reduce the use of parasiticide products and help to increase dung beetle populations.

Tardigrades

Tardigrades, often called water bears or moss piglets, are tiny (0.1–1 mm), eight legged micro animals commonly found in moss, leaf litter, freshwater and soil. In soil they feed on microbes and organic matter, helping decomposition and nutrient cycling within the soil food web. Remarkably they can enter a dried, dormant state called cryptobiosis that allows them to survive extremes, from intense heat or cold, to radiation, and even the vacuum of space.

PRINCIPLE 5

Build or Maintain Soil Organic Matter

Why?

Soil organic matter (SOM) plays a vital role in soil stability by glueing particles together into aggregates and giving the soil a sponge-like quality that helps it resist compaction and maintain porosity[2]. It supplies and retains water and nutrients, stores carbon, and supports complex soil food webs and biological activity. In addition, it reduces soil vulnerability to climatic stress, suppresses soil-borne pests and diseases, buffers pH, and promotes plant growth. The long-term benefits of building up soil organic matter—especially to worn out soils—are endless!

What to do?

- **Keep living roots in the soil** for as many days of the year as possible to build SOM – plants are like big solar panels pumping energy into your soil! Plant growth, powered by photosynthesis, is the main way to move carbon deeper into the soil through root exudates and decaying plant material. Permanent grasslands are excellent at increasing SOM, and the use of cover crops (such as legumes or other green manures, perennial crops, or deep-rooted species) or allowing natural cover regeneration between the main crops may also help to maintain C in tillage systems.
- **Rotate and diversify cropping**. Mixing complementary plant species — whether as multi-species swards in grassland or as diverse crops in a carefully sequenced rotation - provides a wider range of organic inputs, root structures, rooting depths and residue qualities. This improves soil structure and can help build SOM. Incorporating perennial crops (or leys) into arable rotations can also increase SOM.
- **Retain and incorporate straw and other crop residues** into the soil rather than removing them. This practice provides a steady carbon input, reduces soil organic matter decline and supports soil microbial activity that stabilises carbon over time. In wet years, consider the cost of damaging your soil during baling or hauling straw before deciding whether to bale or chop.

Soil Organic Matter

The amount of SOM in your soil is one of the best indicators of soil health, and over half of it is carbon. Irish soils generally already have good levels of SOM, so it's important to protect what we have to help regulate the climate. Building SOM takes time, and changes can be hard to see in the short term, especially in soils that are already rich in SOM. Different types of soil store carbon in different ways. Some soils can build up carbon more easily, while others take more time and effort. The best results come from using several practices together, tailored to the local soil and climatic conditions, to build and maintain SOM over the long term.

- **Maintain and Build SOC with Organic Inputs**. Return organic matter to soil through compost, manure, slurry, crop residues, and other organic amendments. Integrating animal and cropping systems can help replenish SOM, especially in arable systems or on silage blocks where depletion is common. These materials not only supply stable carbon but also improve soil structure, water retention, and nutrient availability, creating conditions that support long-term carbon sequestration.
- **Reduce Soil Disturbance.** Minimise tillage intensity, frequency, and depth in croplands to slow SOM decomposition. Shallow inversion tillage (<17 cm) can balance weed control with carbon retention. Grasslands, which naturally store more SOC (and thus have more to lose!), should be protected from unnecessary disturbance by reducing machinery passages and stocking rates.
- **Limit Soil Compaction & Trafficking.** Reduce heavy machinery use to preserve soil aggregates that protect SOC, while maintaining good soil structure for carbon stabilization.
- **Support Soil Biology**. Encourage healthy soil microbial activity, which stabilises organic matter inputs and enhances long-term carbon storage (principle 4).
- **Prevent Soil Erosion & Degradation**. Protect soils from erosion through conservation practices, ground cover, and reduced disturbance to safeguard stored carbon.
- **Protect Carbon-Rich Soils**. Avoid drainage of organic-rich soils and peatlands, which are highly vulnerable to carbon loss when disturbed. Consider water table management instead on these soils to raise the water table to minimise gaseous emissions.

PRINCIPLE 6

Maintain balanced soil fertility

Why?

Maintaining balanced soil fertility and pH supports healthy crop growth, improves soil health, enhances carbon storage, and reduces input costs. It also supports essential soil functions like nitrogen fixation and residue breakdown, which are vital for long-term productivity.

What to do?

- **Know the soils you have**. Soils differ in their ability to produce crops. Some soils have inherent limits – such as being naturally waterlogged, very shallow, or very acidic/alkaline - that reduce their yield potential. Knowing the type of soil on your farm helps you align production goals with the soil capabilities. On some soils, other functions—like supporting species-rich grasslands or storing carbon—may be more important than growing food. In these cases, managing pH and fertility may be less critical or even undesirable. In moderately or intensively managed soils, where food production is a priority, soil fertility and pH management is essential.
- **Regularly test moderately or intensively managed soils for pH and key nutrients** like phosphorus (P) and potassium (K). It is recommended that intensively managed soils are tested every 2 years.
- **Use a nutrient management plan** to target nutrients where they're most needed on the farm[18]. Follow the 4Rs of nutrient management - use the Right product, at the Right rate, in the Right place, and at the Right time to get the best return and avoid nutrient losses.

Cation Exchange Capacity (CEC)

CEC is the ability of soil particles to attract, hold, and release positively charged ions, including important plant nutrients such as potassium, calcium, magnesium and ammonium. Clay minerals and organic matter largely determine CEC, so soils with high clay or organic matter have higher CEC. CEC is generally a stable property of soil (although it can be gradually influenced by liming or changes in organic matter). You therefore may need to adjust the management to the CEC in soil.

- **Prioritise use of organic manures**, such as slurry or farmyard manure, to reduce the need for chemical fertilisers. Test your slurry to understand its nutrient content and match it with soil test results and crop requirements. Organic manures should be targeted at fields with a requirement for phosphorus (P) and potassium (K) or soils that are low in soil organic matter for example, silage fields and arable. Time your manure application, and use a low emission spreading technique where possible, to get best value from your nutrients.

Maximum nutrient efficiency is usually achieved in spring, when soil temperatures are above 6 °C and rising. Apply organic manures only when the ground is suitable and heavy rain is not expected, to prevent nutrient losses and environmental damage.

- **Promote nitrogen fixation** through use of legumes where appropriate.
- **Aim for balanced soil nutrition** by applying only the nutrients your crops need, considering those supplied naturally through nitrogen fixation and mineralisation processes. Replace nutrients and carbon removed during harvest to maintain long-term soil health. Avoid overloading the soil with any nutrient, as an excess can upset the natural balance, harm the environment, damage beneficial plant–microbe relationships, reduce biodiversity, cause acidification, and affect your bottom line. For example, over-supplying N or P can reduce native N availability or disrupt beneficial mycorrhizal associations that help supply P to plants.
- **Know your soil's capacity to retain and supply nutrients**, which is strongly influenced by CEC and organic matter content. If your soil has low CEC or is peaty or very organic, apply nutrients little and often, and only when the crop needs them, to improve efficiency and uptake. Sandy soils have low CEC and therefore low nutrient-holding capacity. Peaty soils have high CEC and retain nutrients well chemically, but nutrients are often tied up in organic matter and released slowly through mineralisation, so plant-available forms are limited. Some soils can "fix" nutrients, meaning they hold them too tightly for plants to use. This often occurs in acidic soils rich in iron or aluminium or alkaline (limy) soils high in calcium. For these soils, applying nutrients little and often is also the most effective approach
- In fertilised systems **keep soil pH at optimal levels** to ensure nutrients are available, support biological activity including mineralisation and N fixation processes, and prevent soil acidification linked to repeated inorganic fertiliser use. Lime should only be applied based on a recent soil test report and is best applied in autumn while soils are dry and trafficable.

Great soil groups of Ireland

Soil is grouped or classified based on the similarities in their biological, chemical, and physical characteristics[3]. In Ireland, soils are classified at three hierarchal levels (Table 1).

To classify soils, we need to define the different layers (horizons) within the soil profile. In Ireland, Great Soil Groups are the broadest soil classification unit. Soils within each group share similar characteristics and have developed under similar climate and vegetation conditions. These are divided into 56 subgroups based on features relevant to soil management — for example, gleying associated with seasonal or continuous waterlogging. The subgroups are in turn divided into soil series according to dominant texture class and parent material.

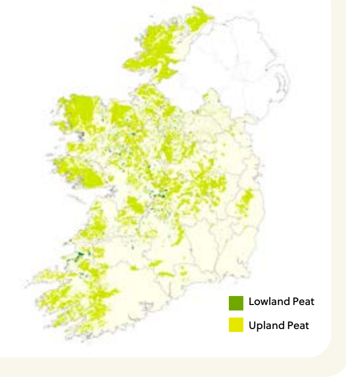
Table 1. Soil classification example at three hierarchal levels from the most general to the most specific. Level 1 = 'Great Soil Groups', Level 2 = 'Subgroups,' and Level 3 = 'Series'

Classification level	Numbers in Ireland	Example					
Great soil group	**11**	**Lithosol**					
Subgroup	**56**	**Typical**		**Histic**		**Humic**	
Soil Series	213	Figar	Glenlane	Kilkieran	Bantry	Knocklyon	Knockshigowna

The following 11 Great Soil Groups have been recorded in Ireland:

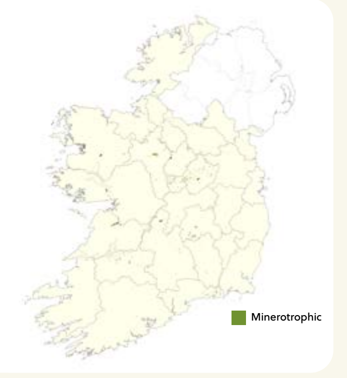
Ombrotrophic peat

- Thick organic (>40cm) layer
- Rain-fed
- pH <4
- Located in lowland (raised bog) or upland (blanket bog) areas



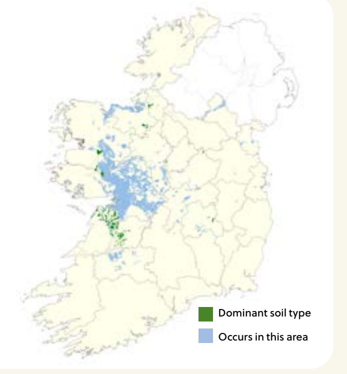
Minerotrophic peat

- Thick organic (>40cm) layer
- Groundwater-fed
- Poor decomposition and pH >4.5
- Located near streams, hollows or other areas with high groundwater table



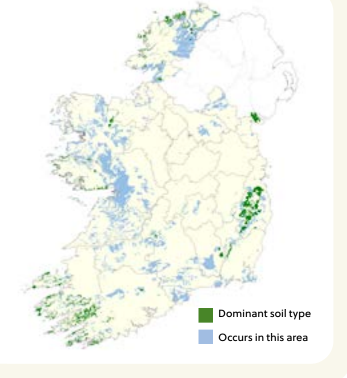
Rendzina

- Shallow <30cm and often very stoney
- Calcareous bedrock; pH>7
- Mainly found in the west of Ireland and Burren



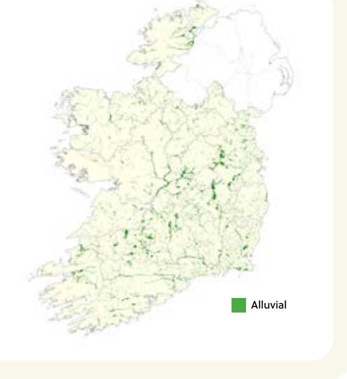
Lithosol

- Shallow <30cm and often very stoney
- Acidic parent material; pH<6.5
- Located in mountain landscapes



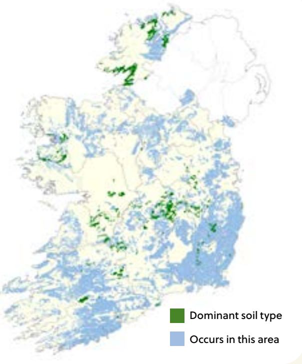
Alluvial

- Forms in floodplains of waterbodies
- Often waterlogged for prolonged periods
- Distinct layers of soil material from deposits
- Found throughout the country



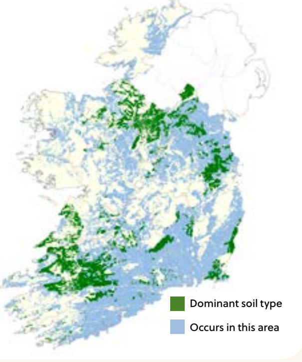
Groundwater gley

- High groundwater table
- Prolonged periods of waterlogging within 40cm of surface
- Grey uniform colour (gleying) or mottling through profile
- Often found near rivers or hollows between drumlins



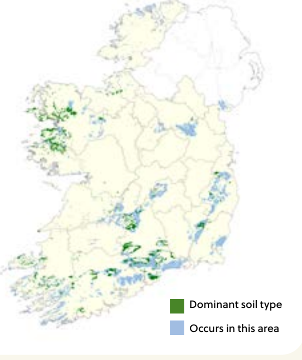
Surface water gley

- Poorly drained with slowly permeable subsurface layer that impedes drainage
- Subsoil dense or with high clay content
- Seasonal prolonged waterlogging of top 40cm causing mottling
- Common in the south west and in border counties



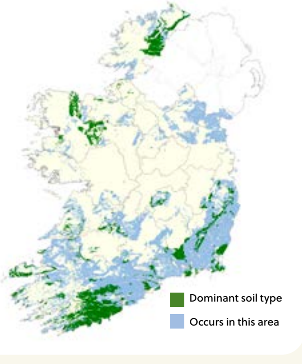
Podzol

- Pale nutrient-poor acidic mineral topsoil overlain with a dark peaty layer
- Organic acids from vegetation result in nutrient leaching (particularly Fe/Al) from surtace layers
- Found at higher altitudes



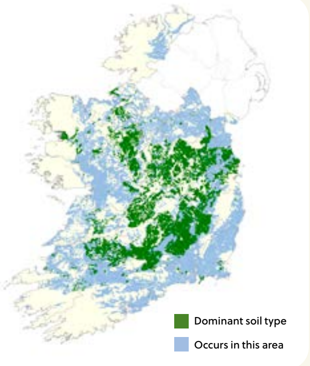
Brown podzolic

- Formed in acidic conditions
- Tends to be free-draining
- Found mainly in the south of the country



Luvisol

- Clay enriched subsoil
- Good agricultural soils with good nutrient retention
- Moderately to imperfectly drained
- Widely distributed



Brown earth

- Young soils with little profile development
- Well drained with good nutrient retention
- Most widely cultivated soils in Ireland

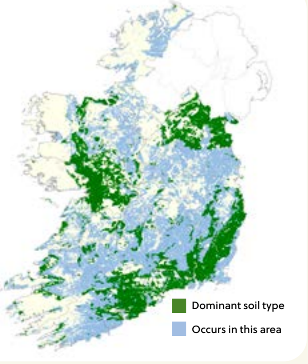


Fig. 9 Soil groups and their typical location within the landscape

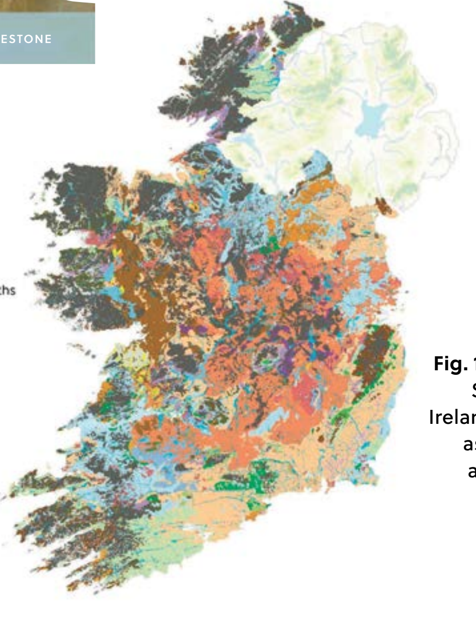


Fig. 10 National Soil map of Ireland with soil associations at a scale of 1:250,000

For the purpose of soil mapping, Ireland's soils are categorised into soil associations, groups of soil series that commonly occur together within a landscape.

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Acknowledgements:

This handbook was designed as part of the Groundtruth project, which was funded through the Research Ireland Discover Programme, and co-funded by the Department of Education. We thank farmers Andrew Bergin and Colm Flynn for their valuable suggestions and constructive feedback on an earlier draft of this text.

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